

Weitong Kong

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GitHub: <https://github.com/MKong17>

EDUCATION

M.Sc. in Mechatronics and Information Technology

Karlsruhe Institute of Technology

Oct. 2022 – Sep. 2025

GPA: 2.3 / 1.0

B.Eng. in Mechanical Engineering

Qingdao University of Science and Technology

Oct. 2015 – Jun. 2020

GPA: 3.6 / 5.0

MASTER THESIS

Domain Generalization under Noisy Labels for LiDAR-based 3D Semantic Segmentation

Nov. 2024 – Aug. 2025 KIT Institute for Anthropomatics and Robotics (IAR)

- Designed a systematic framework to improve robustness of 3D semantic segmentation under label noise and cross-domain shifts.
- Constructed a noise sensitivity benchmark by adapting methods from image-based learning to 3D point clouds.
- Developed a dual-branch architecture with pseudo-label contrastive learning for robust domain generalization.

PUBLICATION

W. Kong*, et al. "Exploring Single Domain Generalization of LiDAR-based Semantic Segmentation under Imperfect Labels", Accepted by IEEE ICRA 2026.

<https://arxiv.org/pdf/2510.09035>

W. Kong*, et al. "IMPACT-CYCLE: A Contract-Based Multi-Agent System for Claim-Level Supervisory Correction of Long-Video Semantic Memory."

<https://arxiv.org/pdf/2604.20136>

*First author

RESEARCH EXPERIENCE

Structured Video Understanding for Panoramic and Stereo Outdoor Hazard Scenes

Dec. 2025 – May. 2026 KIT Institute for Anthropomatics and Robotics (IAR) | Research Assistant (Hiwi)

- Developed a video understanding pipeline for panoramic and stereo scenes with hierarchical semantic representation.
- Built scene-graph representations capturing objects, hazards, and spatial relations in outdoor environments.
- Investigated LVLM video understanding across panoramic and stereo perspectives through comparative analysis.

Language-Aligned 3D Scene Understanding

May. 2026 – Aug. 2026 KIT Institute for Anthropomatics and Robotics (IAR) | Research Assistant (Hiwi)

- Exploring semantic knowledge transfer from Vision-Language Models to 3D point cloud representations for enhanced scene understanding.

TECHNICAL SKILLS

- **Programming:** Python (PyTorch, 3D vision, HPC), working knowledge of C++ and C
- **Languages:** Chinese (Native), English(Intermediate), German (Intermediate)

REFERENCE

Prof. Dr.-Ing. Rainer Stiefelhagen:

Director, Computer Vision for Human-Computer Interaction Lab (cv:hci)

Institute for Anthropomatics and Robotics (IAR) | Karlsruhe Institute of Technology (KIT)